Performance Testing

Date	30 SEPTEMBER 2026
Team ID	SWTID-2026-1660
Project Name	AI WEATHERWISE API
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Weather API / City Weather Search
Test Environment	Local
Test Date	28 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single city weather request	1	60	Response received successfully
2	Multiple weather requests	5	60	API responds correctly
3	Repeated city searches	10	60	Stable response time
4	Invalid city requests	10	60	Proper error message displayed

Step 3: Performance Test Results

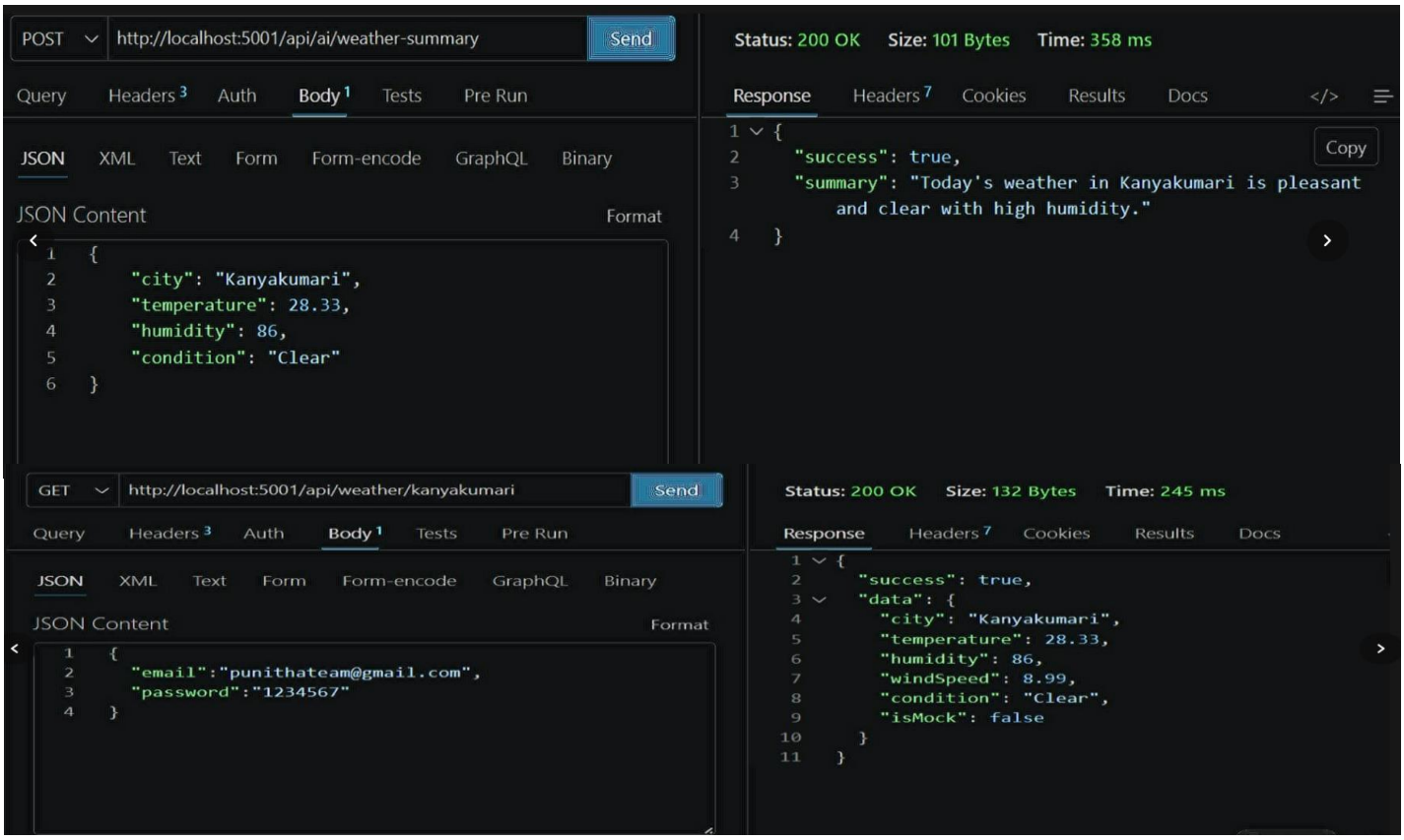
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Record from Postman	Pass/Fail	Based on test
2	Response Time (Max)	< 5 seconds	Record from Postman	Pass/Fail	Based on test
3	Throughput (Req/sec)		Record from test	Pass/Fail	Requests handled
4	Error Rate	< 1%	Record from test	Pass/Fail	API errors checked
5	CPU Utilization	< 80%	N/A	N/A	Local testing
6	Memory Utilization	< 80%	N/A	N/A	Local testing

Step 4: Observations & Analysis

The WeatherWise API was tested using Postman. The API successfully provided weather information for valid city names. Invalid city names displayed an appropriate error message. The system performed well during testing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


POST
http://localhost:5001/api/ai/weather-summary
Send
Query
Headers 3
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
{
"city": "Kanyakumari",
"temperature": 28.33,
"humidity": 86,
"condition": "Clear"
}
Status: 200 OK
Size: 101 Bytes
Time: 358 ms
Response
Headers 7
Cookies
Results
Docs
Copy
{
"success": true,
"summary": "Today's weather in Kanyakumari is pleasant and clear with high humidity."
}

GET
http://localhost:5001/api/weather/kanyakumari
Send
Query
Headers 3
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
{
"email":"punithateam@gmail.com",
"password":"1234567"
}
Status: 200 OK
Size: 132 Bytes
Time: 245 ms
Response
Headers 7
Cookies
Results
Docs
{
"success": true,
"data": {
"city": "Kanyakumari",
"temperature": 28.33,
"humidity": 86,
"windSpeed": 8.99,
"condition": "Clear",
"isMock": false
}
}